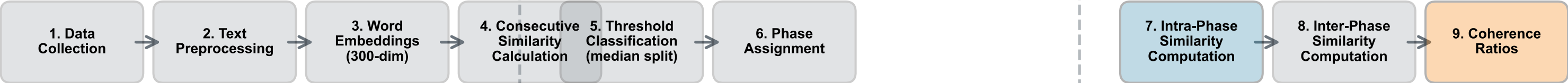


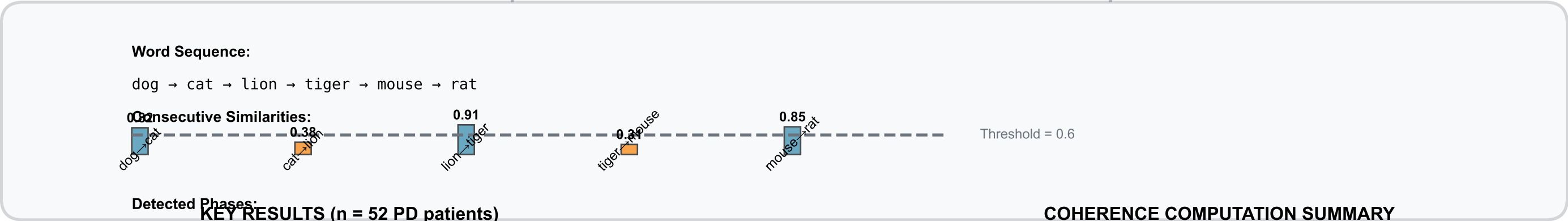
SECTION 1: DATA PROCESSING

SECTION 2: PHASE DETECTION

SECTION 3: COHERENCE COMPUTATION



DETAILED EXAMPLE: Phase Detection



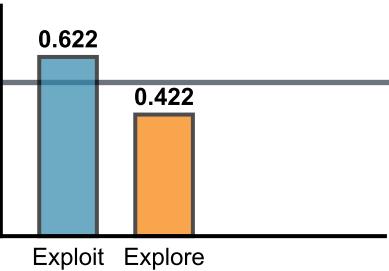
KEY RESULTS (n = 52 PD patients)

COHERENCE COMPUTATION SUMMARY

- Exploitation: [dog, cat], [lion, tiger], [mouse, rat]
- Exploration: [cat→lion], [tiger→mouse]

Intra-Phase Coherence

- Exploitation coherence = 0.622 ± 0.146
- Exploration coherence = 0.422 ± 0.083
- E-E index = 1.52 ± 0.39 (75% > 1.0)
- Paired t-test: $t = 8.83$, $p < 0.001$



- Intra-Phase Similarity:**
- Pairwise similarities within each phase type
 - Separate means for exploitation and exploration phases

- Inter-Phase Similarity:**
- Centroid-based similarity between all phases
 - Single mean across all phase pairs

- Final Ratios:**
- Exploitation Coherence = $\text{Intra_exploit} / \text{Inter}$
 - Exploration Coherence = $\text{Intra_explore} / \text{Inter}$